

RESEARCH ARTICLE



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Interface-engineered polypropylene/cellulose nanofibril composites with enhanced thermal stability, mechanical modulus, and impact strength

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Abstract

To attain interface-engineered polypropylene/cellulose nanofibril (PP/CNF) composites with improved thermal and mechanical performance, PP was grafted with glycidyl methacrylate (GMA) and styrene via direct melt-mixing and GMA/styrene-grafted PP (PGMA) was reinforced with 1–10 wt% CNF content through a mass-producible melt-compounding process. The spectroscopic analyses confirmed the existence of specific hydrogen-bonding interactions between CNF and PGMA in the composites. The melt-rheological measurement revealed the occurrence of chemical reactions between the hydroxy group of CNF and the glycidyl group of PGMA of the composites. Accordingly, the dispersion and interfacial adhesion of CNF in the PGMA matrix was improved. β -form PP crystals were also developed by accelerating the crystallization of PGMA and PGMA/CNF composites, unlike PP with only α -form crystals. The storage modulus of PGMA/CNF composites was enhanced noticeably with increasing CNF content. Moreover, the impact strength of PGMA composite with 1 wt% CNF was found to be 29.2% higher than that of neat PP.

KEYWORDS

cellulose nanofibril, mechanical properties, polymer composites, polypropylene, thermal properties

1 | INTRODUCTION

Isotactic polypropylene (PP), the world's second largest petroleum-based thermoplastic polymer since 1957, possesses a lot of preferable properties such as low cost, low density, recyclability, easy processing, and high thermal stability.^[1] Accordingly, PP has been used in a wide range of applications such as packaging, textile, stationery, reusable containers, laboratory equipment, automotive components, and so forth. The mechanical properties of PP are between those of commodity and engineering thermoplastics. The properties of PP can be tailored through different reinforcements to improve its performance and expand its applications. In addition, to

overcome environmental pollution and global warming, thermoplastic polymer composites reinforced with natural or renewable fillers such as cellulose nanocrystals, cellulose nanofibrils (CNFs), nanoclays, natural fibers, and so forth have gained lots of attention from industry and academia.^[2–8]

Among those renewable nanofillers, CNFs with a diameter of 5–30 nm and an aspect ratio greater than 50, derived from natural cellulosic sources by mechanical disintegration, chemical and enzymatic treatment, have attracted great interest from industry and academia due to their lightweight, nontoxic, recyclability, and biodegradability. Accordingly, PP/CNF composites have been extensively investigated. However, unlike CNFs with a highly polar surface,^[9] PP is hydrophobic without any

polar groups in its backbone.^[10,11] The main problem encountered with PP/CNF composites stems from the incompatibility between the hydrophilic CNF and the hydrophobic PP matrix. This may contribute to a poor stress transfer between the matrix and the filler.^[12–17]

Wang et al.^[15] investigated the morphology and process-property relationship of PP nanocomposites containing CNFs with different degrees of substitution (surface modification) using hydrophobic alkenyl succinic anhydride. The hydrophobically surface-modified CNFs were found to be dispersed uniformly in the PP matrix as well as to improve significantly the crystallization kinetics and foamability of PP, compared with unmodified CNF. Peng et al.^[14] report the thermal and mechanical properties of PP composite films reinforced with micro-sized CNFs containing different lignin contents. They found that high lignin content (>20 wt%) CNFs showed better dispersion in the PP matrix than low lignin content (<5 wt%) CNFs and that significantly enhanced thermal stability, moisture resistance, and tensile strength/modulus were attained for PP composite films with high lignin content CNFs.

Numerous studies have been carried out for improving the interfacial bonding between cellulosic fibers and polymer matrices through the surface modification of CNFs including noncovalent modifications using surfactants and polyelectrolytes or covalent modifications using polymer grafting, silylation, cationization, esterification, and so forth.^[18–24] Those methods are effective to obtain hydrophobic surfaces but are limited in aspects of fabrication cost, environmental concern, and complex reaction conditions. Instead, grafting glycidyl methacrylate (GMA) and styrene onto PP chains is a simple and efficient method to endow the hydrophilic functionality of PP.^[25] In addition, it is expected that the epoxy and ester groups of GMA/styrene-grafted PP (PGMA) can chemically react and physically interact with the hydroxyl groups of CNF, respectively, thereby achieving improved interfacial bonding between CNF and PGMA matrix.

In this study, a series of PGMA-based composites possessing 1–10 wt% CNF loadings were prepared by masterbatch melt-compounding followed by injection molding. For this purpose, PGMA was prepared via reactive melt-mixing of neat PP in the presence of GMA, styrene, and radical catalyst. The successful fabrication of PGMA and the interfacial interactions of PGMA/CNF composites at a molecular level was analyzed using Fourier transform infrared (FT-IR) spectroscopy and energy dispersive X-ray spectroscopy (EDS). The morphological and crystalline features of PGMA/CNF composites were identified using scanning electron microscopy (SEM) and X-ray diffraction method (XRD), respectively. The rheological, thermal, and mechanical properties of the composites were measured using melt rheometer technique differential scanning calorimetry (DSC),

thermogravimetric analysis (TGA), dynamic mechanical analysis (DMA), and Izod impact test. The thermal and mechanical performance of PGMA/CNF composites was analyzed by taking into account CNF content, interfacial interaction, and grafting reactions in the composites.

2 | EXPERIMENTAL SECTION

2.1 | Materials

Isotactic PP (Mn of 250,000 g/mol and Mw of 670,000 g/mol), GMA (97%), styrene (>99%), and benzoyl peroxide (BPO) were purchased from Sigma-Aldrich Com. CNF having a width of 80–300 nm, a length of 100–2000 μm , an aspect ratio of >1000, and a specific surface of >80 m^2/g , which was manufactured by stripping filaments from wood fibers through a mechanical process suitable for mass production without chemical or enzymatic treatment, was obtained from Kruger Biomaterials Inc.

2.2 | Preparation of PGMA and PGMA/CNF composites

GMA/styrene-grafted PP (PGMA) was fabricated by reactive melt-compounding of neat PP chips in the presence of GMA, styrene, and BPO initiator using a corotating intermeshing twin-screw extruder (BA11, Bautek Com.) set at a rotor speed of 60 rpm and a temperature profile of 175–195°C.^[25] Prior to melt-processing, a solid mixture of 550 g PP powder, 10 wt% GMA, 10 wt% styrene, and 0.2 wt% BPO was prepared. Styrene was adopted to increase the grafting efficiency of GMA on PP chains. The solid mixture was then melt-compounded through the twin-screw extruder. The unreacted GMA and styrene components of the melt-compounded PGMA extrudate were removed by dissolving the extrudate in xylene at 120°C and precipitating it in acetone at 25°C. The purified PGMA precipitate was then obtained via filtering, washing with acetone, and drying at 60°C under vacuum.

A series of PGMA/CNF composites were manufactured through masterbatch melt-compounding, as represented schematically in Figure 1. To minimize the effect of moisture during melt-compounding, PGMA and CNFs were vacuum-dried at 60°C for 24 h. To improve the compatibility of CNFs in the PGMA matrix, a masterbatch consisting of 90 wt% PGMA and 10 wt% CNF was manufactured by melt-mixing through the twin-screw extruder set at 60 rpm and 175–195°C. Finally, PGMA/CNF composites with 1–10 wt% CNF content were fabricated by melt-mixing the masterbatch with neat PGMA, cutting into pellets, and drying in a vacuum oven at 60°C. The melt-compounded composites were

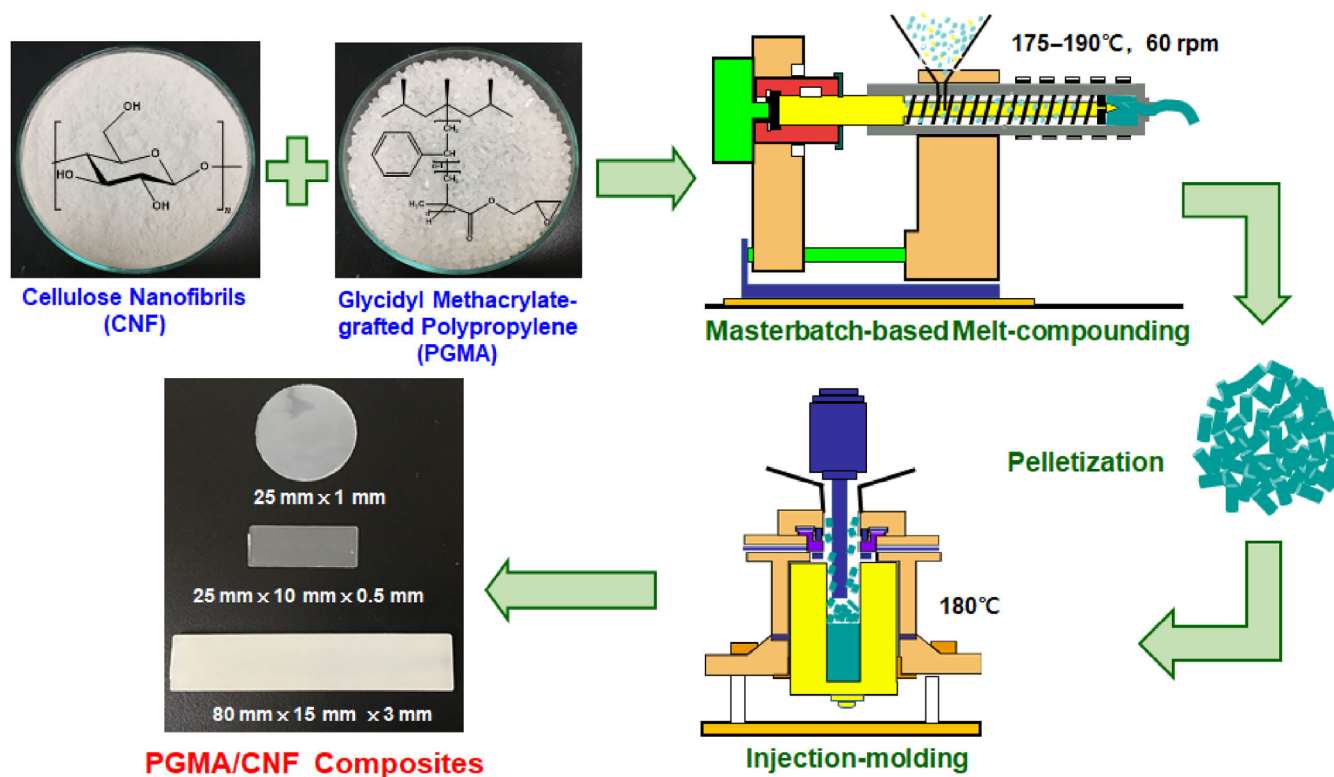


FIGURE 1 Schematic procedure for preparing PGMA/CNF composites through melt-compounding and injection-molding

TABLE 1 Relative composition of PGMA/CNF and PP/CNF composites

Sample code	PGMA (wt%)	CNF (wt%)	PP (wt%)
PGMA/CNF1	99	1	0
PGMA/CNF3	97	3	0
PGMA/CNF5	95	5	0
PGMA/CNF7	93	7	0
PGMA/CNF10	90	10	0
PP/CNF10	0	10	90

named PGMA/CNF_x, where *x* represents the loading content of CNF in wt%, as listed in Table 1. For comparison, PP composite with 10 wt% CNF (PP/CNF10) was manufactured by the same manufacturing procedure. For structure and property characterization, PGMA/CNF and PP/CNF composites were molded into specimens with three different dimensions through an injection molding machine (M-18, QNES Global Com.). The injection molding temperature was kept at 180°C.

2.3 | Characterization

The chemical modification of neat PP and the degree of GMA grafting were characterized using an X-ray

photoelectron spectrometer (XPS, K-Alpha, ThermoFisher Scientific Com.) with a monochromatic radiation source of Al-K α .

The molecular-level structures of neat PP, PGMA, CNF, and their composites were characterized using a FT-IR spectrometer (Nicolet iS10, ThermoFisher Scientific Com.).

The morphological and elemental mapping analyses of fractured surfaces of PGMA/CNF composites were carried out with the aid of a SEM (CLARA, TESCAN Com.) coupled with an energy dispersive EDS (Ultim Max, Oxford Instruments Com.).

The crystalline features of composites were analyzed using a wide-angle XRD (D8 DISCOVER, Bruker AXS Com.) in reflection mode with Cu-K α radiation.

The thermal transition characterization of composites was performed by a DSC (DSC6000, PerkinElmer Inc.). DSC heating and cooling thermograms were obtained under constant nitrogen gas flow over a temperature range of 30–200°C at a rate of 10°C/min.

The thermal degradation and stability of PGMA/CNF composites was analyzed using a TGA (TGA4000, PerkinElmer Inc.). TGA thermograms were obtained under an inert nitrogen environment over a temperature range of 30–600°C with a heating rate of 10°C/min.

The temperature-dependent mechanical properties of composites were investigated with a DMA (DMA8000,

PerkinElmer Inc.). The DMA tests in single cantilever mode were performed at an applied frequency of 0.1 Hz at temperatures from -50 to 150°C with a ramp rate of $3^{\circ}\text{C}/\text{min}$.

The melt-rheological properties of PGMA/CNF composites were examined using a rheometer (MCR 102, Anton Paar Com.) equipped with parallel plates with a diameter of 25.0 mm and a gap of 1.0 mm. The melt viscosity of the composites at various temperatures of 180, 200, and 220°C was monitored as a function of time under a constant shear rate of 0.05 s^{-1} .

The impact strength of PGMA/CNF composites was measured by an Izod impact tester (QM750A, QMESYS Com.). For the Izod impact test, all samples measuring $80\text{ mm} \times 15\text{ mm} \times 3\text{ mm}$ were machined with a V-shaped notch using a mechanical cutter. Five tests were performed to report average values and deviations of impact strength for each composite sample.

3 | RESULTS AND DISCUSSION

3.1 | Microstructure and morphology

The XPS spectra and elemental composition of neat PP and PGMA were displayed in Figure 2. From the XPS

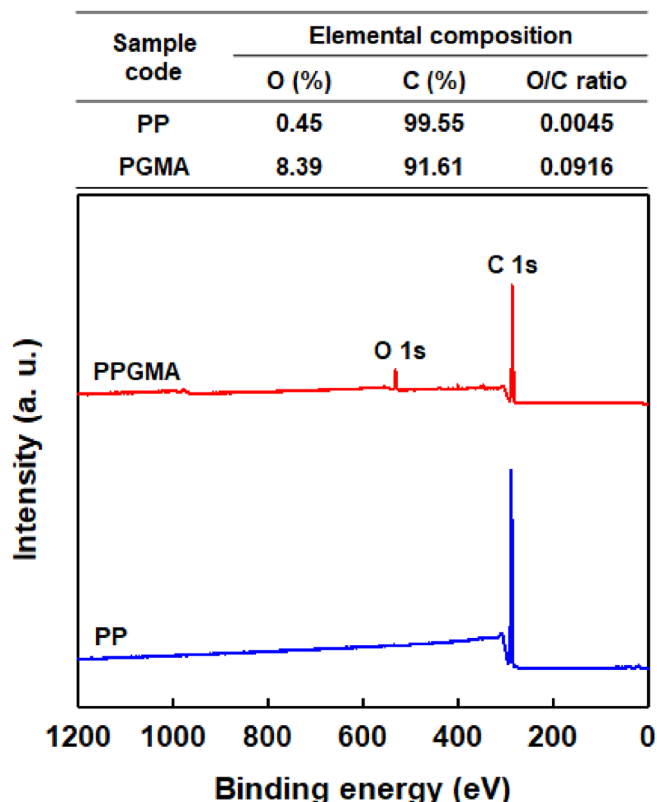


FIGURE 2 XPS survey spectra and elemental composition of FT-IR spectra of neat PP and PGMA

survey spectra, the elemental composition of neat PP was evaluated to be 99.55% carbon and 0.45% oxygen. In the case of PGMA, the carbon content decreased to 91.61%, whereas the oxygen content increased to 8.39%, indicating that GMA was successfully grafted onto the PP backbone. In addition, the grafting degree of GMA to the PP backbone can be calculated to be about 11.6%, based on the O/C ratio of PGMA.

To characterize the molecular structures and interactions of PGMA/CNF composites and their components (CNF, GMA, PP, and PGMA), FT-IR spectra were obtained as represented in Figure 3. The characteristic vibration bands were also summarized in Table 2. In the case of pristine CNF, characteristic vibrational bands of hydroxyl ($\nu_{\text{O-H}}$) and methyl ($\nu_{\text{C-H}}$) groups were detected at ~ 3300 and $\sim 2900\text{ cm}^{-1}$, respectively. Neat PP shows the typical stretching ($\nu_{\text{C-H}}$) and bending ($\delta_{\text{C-H}}$) vibrational bands of $=\text{CH}-$, $-\text{CH}_2-$, and $-\text{CH}_3$ groups at ~ 2917 and $\sim 1454\text{ cm}^{-1}$, respectively. The FT-IR spectrum of PGMA is almost identical to that of neat PP, except for the appearance of new bands at ~ 702 and $\sim 1726\text{ cm}^{-1}$. The absorption band at $\sim 702\text{ cm}^{-1}$ is owing to the C-H out-of-plane bending vibration ($\gamma_{\text{C-H}}$) of the phenyl group of styrene,^[26] while the band at $\sim 1726\text{ cm}^{-1}$ is due to the carbonyl stretching vibration ($\nu_{\text{C=O}}$) of the GMA group of PGMA.^[18,27,28] The existence of $\nu_{\text{C=O}}$ band at $\sim 1726\text{ cm}^{-1}$ and the disappearance of $\nu_{\text{C=C}}$ at $\sim 1637\text{ cm}^{-1}$ and $\delta_{\text{C=C}}$ at $\sim 945\text{ cm}^{-1}$ of the GMA group in the spectrum of PGMA show that GMA was successfully grafted on PP chains.^[25,29] Overall FT-IR spectra of PGMA/CNF composites were found to be consistent with the spectrum of PGMA. Nonetheless, the $\nu_{\text{C=O}}$ band was observed to shift from $\sim 1726\text{ cm}^{-1}$ for PGMA to $\sim 1738\text{ cm}^{-1}$ for PGMA/CNF10 composite. It indicates the existence of a strong hydrogen-bonding interaction between the hydroxyl group of CNF and the ester group of PGMA in the composites.

To investigate the dispersity of CNFs in the composites and the interfacial adhesion and compatibility between CNF and PGMA, SEM images of cryogenically fractured surfaces of neat PP, PGMA, and their composites with different CNF loadings were obtained as can be seen in Figure 4. The fracture surfaces of neat PP and PGMA were observed to be rather rough owing to their semicrystalline tough failure (Figure 4A,B). The fracture surfaces of PGMA/CNF composites showed a quite similar morphology to the one of PGMA without showing noticeable CNFs and CNF/matrix boundary (Figure 4C-E). It indicates that CNFs are tightly wrapped by the PGMA matrix, which is due to the specific interactions between PGMA and CNFs in the composites, as supported by the FT-IR analysis. For the PP/CNF10 composite (Figure 4F), CNF fillers and voids exist over the fracture surface owing to the pull-out of CNFs from the

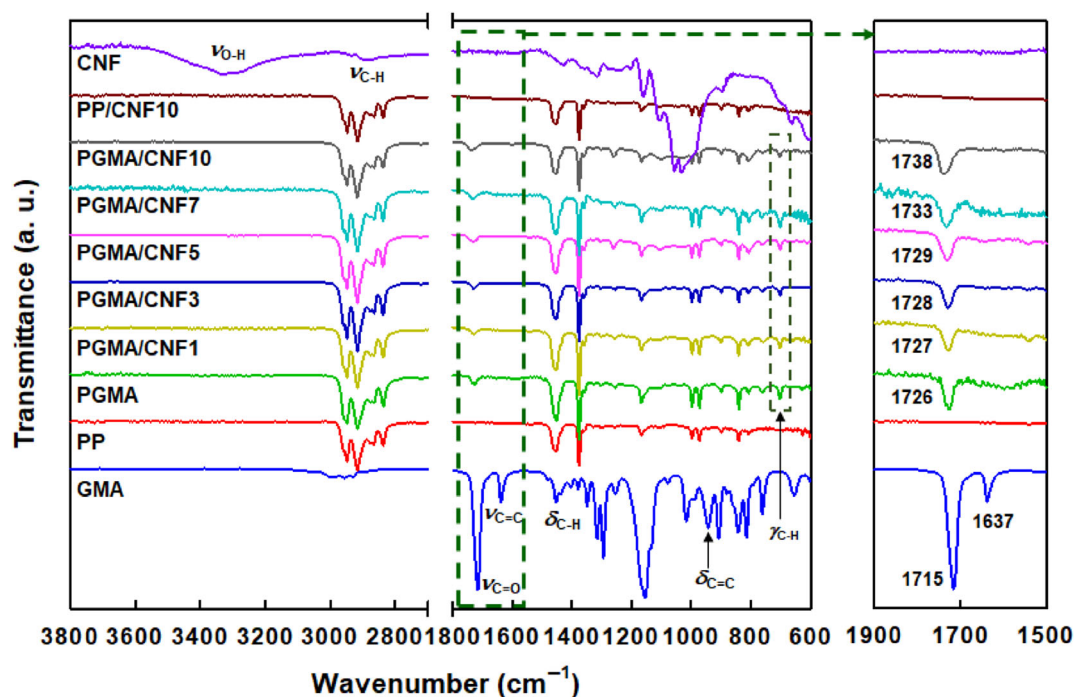


FIGURE 3 FT-IR spectra of PP, GMA, PGMA, CNF, and their composites

TABLE 2 Characteristic vibrational bands of PP, GMA, PGMA, and their composites

Sample code	Vibrational bands (cm ⁻¹)					
	ν_{C-H}	$\nu_{C=O}$	$\nu_{C=C}$	δ_{C-H}	$\delta_{C=C}$	γ_{C-H}
PP	2917	–	–	1454	–	–
GMA	2957	1715	1637	–	945	–
PGMA	2917	1726	–	1452	–	702
PGMA/CNF1	2917	1727	–	1454	–	702
PGMA/CNF3	2917	1728	–	1453	–	702
PGMA/CNF5	2917	1729	–	1453	–	702
PGMA/CNF7	2917	1733	–	1446	–	702
PGMA/CNF10	2917	1738	–	1444	–	702
PP/CNF10	2917	–	–	1453	–	–

PP matrix, which results from the immiscibility between hydrophilic CNFs and hydrophobic PP matrix.

The elemental composition from the fracture surfaces of PP, PPGMA, and their composites was analyzed using the EDS data and the results are represented with the SEM images in Figure 4. Unlike the neat PP, the presence of elemental oxygen in PGMA indicates the successful grafting of GMA onto PP hydrocarbon chains. The higher elemental oxygen content of the PGMA/CNF10 composite, compared with PGMA, originates from the addition of CNFs to PGMA.

3.2 | Rheological properties

The rheological analysis is a useful technique to characterize the processability and microstructure of polymer materials and thermoplastic composites in a melt state. For comparison of the melt-rheological behavior of PGMA/CNF10 and PP/CNF10 composites, the viscosity changes at various temperatures of 180–220°C were monitored as represented in Figure 5. The viscosities of the PGMA/CNF10 composite were found to be higher than those of the PP/CNF10 composite at all temperatures. In addition, the gap in the viscosities between PGMA/CNF10 and PP/CNF10 composites increased from ~52% to ~3543%, as the temperature increased from 180°C to 220°C. This result is dominated by the existence of strong hydrogen bonds between CNF and PGMA matrix in PGMA/CNF composites, as supported by the above FT-IR spectroscopic analysis. It is also remarkable that the melt viscosity of the PGMA/CNF10 composite at 220°C increased gradually with time, while the viscosity of the PP/CNF10 composite decreased. These time-dependent viscosity changes of PGMA/CNF composite are associated with the morphology change as well as the reactivity. Accordingly, the decrement in melt-viscosities of PP/CNF10 composite at 220°C with time stems from the fact that the dispersion of CNFs in the PP matrix becomes poor with time because of no specific interactions between hydrophilic CNFs and hydrophobic PP matrix. On the other hand, the increment in viscosities of PGMA/CNF10 composite at

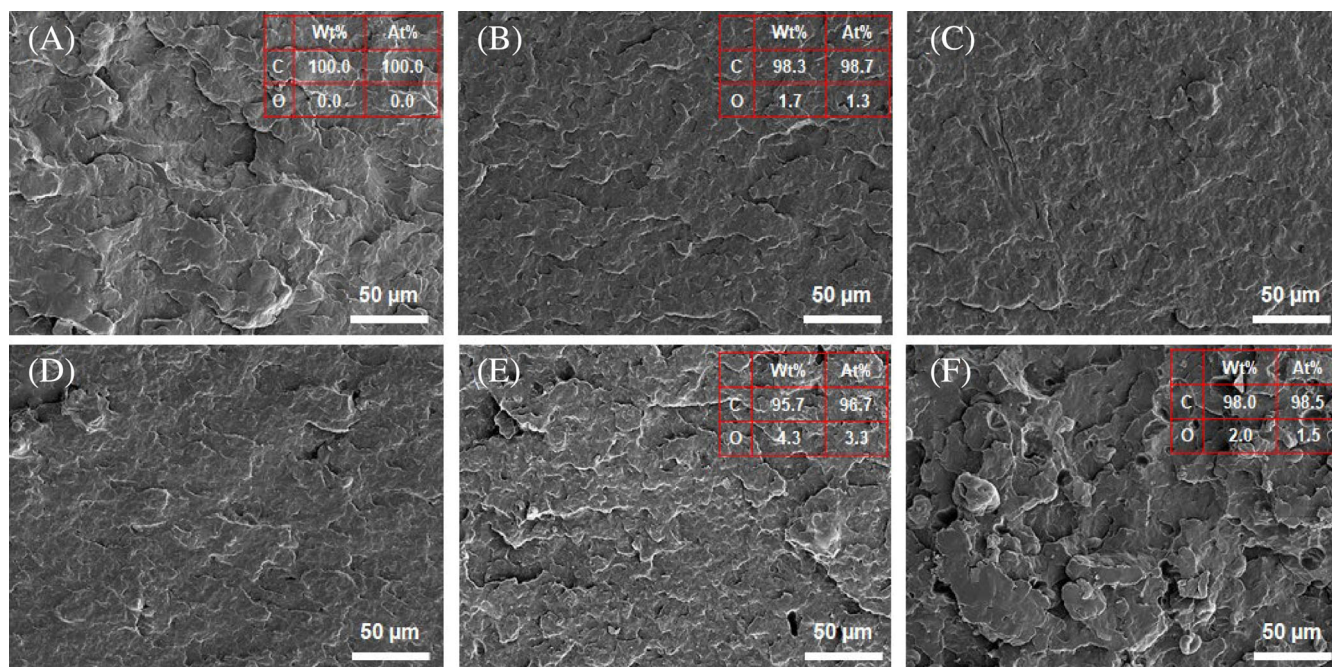


FIGURE 4 SEM images of fractured surfaces of (A) PP, (B) PGMA, (C) PGMA/CNF1, (D) PGMA/CNF5, (E) PGMA/CNF10, and (F) PP/CNF10

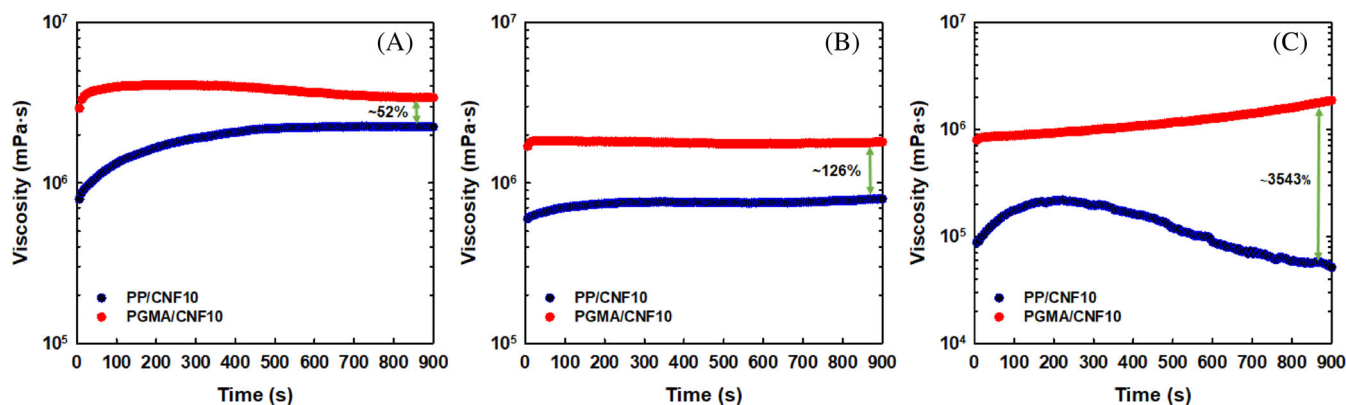


FIGURE 5 Time-dependent melt-viscosity changes of PGMA/CNF10 and PP/CNF10 composites at different constant temperatures of (A) 180°C; (B) 200°C; (C) 220°C

220°C stems from the enhanced compatibility and dispersity owing to the grafting reaction between the hydroxy group of CNF and the epoxy group of PGMA matrix during the measurement. This rheological result is consistent with the above SEM analysis in that CNFs are not identified clearly in the PGMA matrix in the fractured surfaces of PGMA/CNF composites. Overall, from the above FT-IR, SEM, and melt-rheological analyses, it is reasonable to contend that there are specific interactions as well as grafting reactions between the hydroxyl group of CNF and the GMA group of PGMA at the interface of the composites, as schematically represented in Figure 6.

3.3 | Crystalline structure

To identify the crystalline features of PGMA/CNF composites, the XRD patterns of neat PP, PGMA, CNF, and their composites were obtained, as shown in Figure 7. In the case of the CNF component, characteristic diffraction peaks were detected at $\sim 15.0^\circ$, $\sim 16.2^\circ$, $\sim 22.5^\circ$, and $\sim 34.6^\circ$, which correspond to the reflections of (001), (101), (002), and (040) planes of cellulose type I crystals, respectively.^[30] Neat PP displayed characteristic diffraction peaks at $\sim 14.0^\circ$, $\sim 16.8^\circ$, $\sim 18.6^\circ$, $\sim 21.2^\circ$, $\sim 21.9^\circ$, and $\sim 25.6^\circ$, which are associated with the reflections of (110), (040), (130), (111), (041), and (060) planes of α -form

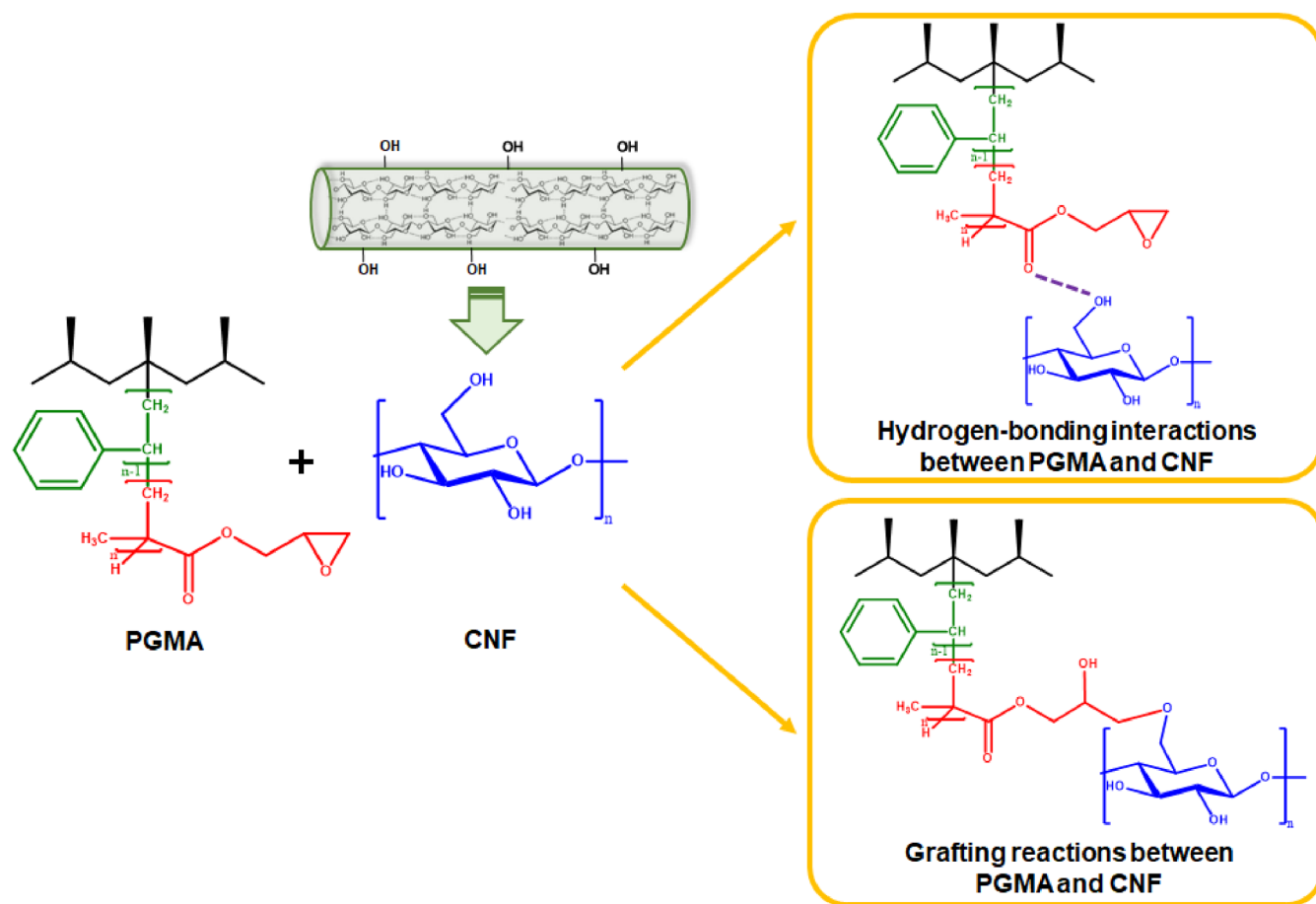


FIGURE 6 Scheme for intermolecular hydrogen-bonding interactions and grafting reactions between PGMA and CNF in the PGMA/CNF composites

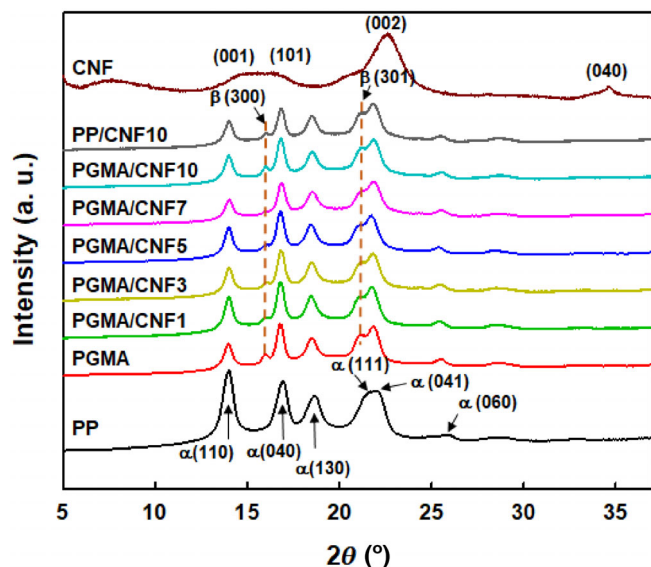


FIGURE 7 X-ray diffraction patterns of PP, PGMA, CNF, and their composites with different CNF contents

crystals, respectively.^[31,32] After adding CNFs into the PP matrix or grafting the GMA onto PP chains, new distinct diffraction peaks at $\sim 15.9^\circ$ and $\sim 20.9^\circ$ appeared and they

originated from the reflections of (300) and (301) planes of the β -form PP crystal, respectively.^[33,34] The characteristic diffraction peaks of the β -form PP crystal were also observed for PGMA/CNF and PP/CNF composites.^[32]

3.4 | Thermal properties

The thermal transition behavior of PGMA/CNF composites was investigated using DSC cooling and heating curves of PP, PGMA, PP/CNF, and PGMA/CNF composites, as can be seen in Figure 8. Typical thermal transition temperatures associated with crystallization and melting are listed in Table 3. In the DSC cooling thermograms (Figure 8A), the melt-crystallization temperature ($T_{mc} \sim 123.8^\circ\text{C}$) of PGMA was found to be higher than that ($\sim 116.2^\circ\text{C}$) of neat PP. In addition, the T_c 's of PGMA/CNF and PP/CNF composites were higher than the values of neat PGMA and PP, respectively. These results suggest that the melt-crystallization rates of PGMA are faster than that of neat PP due to the specific intermolecular interactions among GMA groups grafted on PP chains and that CNF serves as a nucleating agent

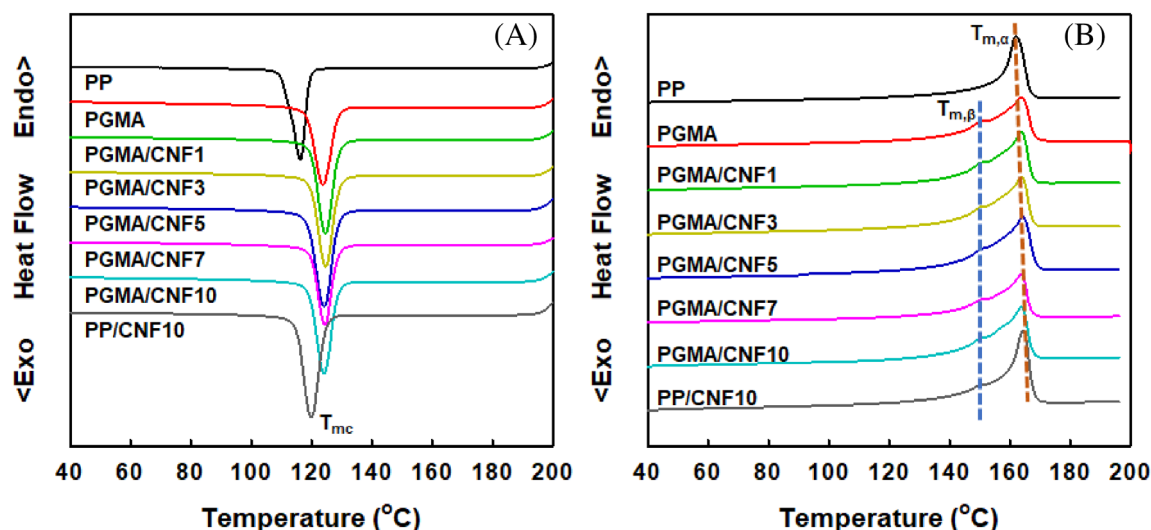


FIGURE 8 DSC (A) cooling and (B) heating thermograms of PP, PGMA, and their composites with different CNF contents

TABLE 3 Thermal transition, decomposition, and loss tangent ($\tan \delta$) peak temperatures of PP, PGMA, and their composites containing different CNF content

Sample code	DSC			TG/DTG		DMA	
	T_{mc} (°C)	$T_{m,\alpha}$ (°C)	$T_{m,\beta}$ (°C)	$T_{d,CNF}$ (°C)	$T_{d,PP}$ (°C)	$T_{\beta,PP}$ (°C)	$T_{\alpha,PP}$ (°C)
CNF	–	–	–	316.4	–	–	–
PP	116.2	162.0	–	–	453.6	14.8	88.0
PGMA	123.8	163.8	150.2	–	435.1	11.7	84.2
PGMA/CNF1	124.7	164.1	150.1	310.5	458.5	13.3	84.5
PGMA/CNF3	124.7	164.1	150.1	304.1	457.9	13.4	84.4
PGMA/CNF5	124.3	164.4	150.4	300.0	458.0	13.9	81.5
PGMA/CNF7	124.5	163.8	150.3	299.3	458.9	13.6	79.5
PGMA/CNF10	124.2	163.9	150.2	297.5	461.5	13.8	78.5
PP/CNF10	120.0	164.5	149.5	300.5	445.1	16.1	83.0

during the melt-crystallization of neat PP and PGMA matrices in the composites.^[25,32,35,36] In the DSC heating thermograms (Figure 8B), PGMA and PGMA/CNF composites exhibited two melting transition temperatures ($T_{m,\alpha}$ and $T_{m,\beta}$), while neat PP shows only one melting transition ($T_{m,\alpha}$). The strong melting transition at a higher temperature ($T_{m,\alpha}$) of $\sim 164^\circ\text{C}$ is related to the melting of the α -form PP crystal, whereas the relatively weak melting transition at a lower temperature ($T_{m,\beta}$) of $\sim 150^\circ\text{C}$ is associated with the melting of the β -form crystal.^[32,37–40] The strong $T_{m,\alpha}$ and weak $T_{m,\alpha}$ in the melting transition are consistent with the relative diffraction intensities of α - and β -form crystal phases in the XRD data of the PGMA and PGMA/CNF composites.

TG and DTG analyses were employed to investigate the effect of CNF content on the thermal decomposition and stability of PGMA/CNF composites, as can be seen

in Figure 9. PP, PGMA, and CNF show one-step thermal degradation behavior. On the other hand, PP/CNF10 and PGMA/CNF composites exhibit two-step mass losses owing to each thermal decomposition of CNF and PGMA (or PP) components. The thermal decomposition peak temperatures ($T_{d,CNF}$ and $T_{d,PP}$) of CNF and PGMA (or PP) components in the composites were evaluated from the DTG curves, as summarized in Table 3. The $T_{d,PP}$ value ($\sim 435.1^\circ\text{C}$) of PGMA was found to be $\sim 18.5^\circ\text{C}$ lower than that ($\sim 453.6^\circ\text{C}$) of the neat PP, which is due to the thermally less stable GMA group of PGMA. When 10 wt% CNF was added to the PP matrix, $T_{d,CNF}$ and $T_{d,PP}$ of PP/CNF10 composite were lower than that of raw CNF and PP components, respectively, due to the incompatibility between hydrophilic CNF and hydrophobic PP in the composite. In the cases of PGMA/CNF composites, the $T_{d,CNF}$ value decreased from 310.5 to 297.5°C

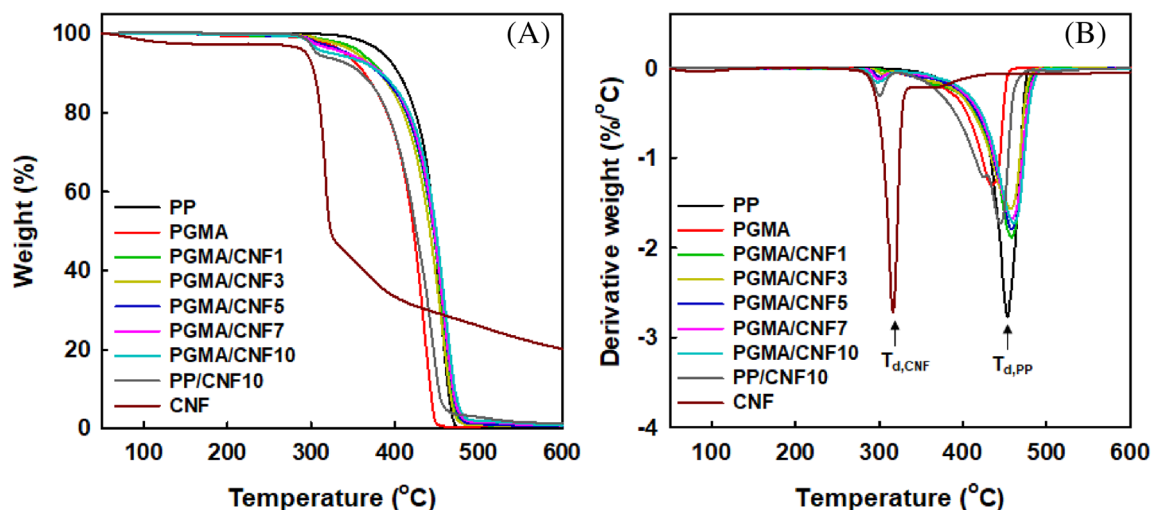


FIGURE 9 (A) TG and (B) DTG thermograms of PP, PGMA, and their composites with different CNF contents

with the increase in CNF content from 1 wt% to 10 wt%, whereas the $T_{d,PP}$ values of 458.5–461.5°C were far higher than neat PGMA (435.1°C) or even neat PP (453.6°C). On the other hand, $T_{d,CNF}$ (300.5°C) and $T_{d,PP}$ (455.1°C) of PP/CNF10 composite were measured to be lower than those (316.4 and 453.6°C) of neat CNF and PP components, respectively. It is thus considered that the noticeably enhanced $T_{d,PP}$ values of PGMA/CNF composites is due to the presence of specific intermolecular interactions as well as chemical reactions between CNF fillers and PGMA matrix, unlike PP/CNF composite without specific interaction and good dispersion.

3.5 | Mechanical properties

The dynamic mechanical thermal analysis is most useful for investigating the viscoelastic behavior of polymeric materials and composites in a solid state. The changes in storage modulus (E') and loss tangent ($\tan \delta$) of PGMA/CNF and PP/CNF composites with time are represented in Figure 10. For PGMA/CNF composites, the storage modulus increased with increasing CNF content (Figure 10A), which is owing to the synergistic reinforcing effect of good dispersion of compatibilized CNFs and effective interfacial stress transfer from the PGMA matrix to the CNF filler.^[41] Furthermore, the storage moduli of PGMA/CNF10 composite were much higher than those of PP/CNF10 composite. The remarkable increment in storage modulus for PGMA/CNF10 composite can be attributed to the enhanced compatibilization and interfacial adhesion by strong hydrogen-bonding interactions and grafting reactions between PGMA and CNF.

Figure 10B shows the changes of loss tangent ($\tan \delta$) with temperature for PP, PGMA, and their composites

with different CNF contents. All components and composites showed two relaxation peaks, α and β . The β relaxation peak ($T_{\beta,PP}$) detected in the range of 10–20°C is associated with the glass transition of PP chains in the unrestricted amorphous phase, while the α relaxation peak ($T_{\alpha,PP}$) observed in the range of 70–90°C is connected with lamellar slip and chain rotation in the restricted crystalline phase.^[42] The $T_{\alpha,PP}$ and $T_{\beta,PP}$ values evaluated from Figure 10B are listed in Table 3. The $T_{\alpha,PP}$ and $T_{\beta,PP}$ values of PGMA were found to be slightly lower than those of neat PP, which is related to the enhanced chain mobility of PP chain segments by grafted GMA groups.^[25] For PGMA/CNF composites, $T_{\beta,PP}$ values of 13.3–13.9°C were higher than that (11.7°C) of neat PGMA, which is owing to the restricted chain mobility of the PGMA matrix interacting with CNFs in the composites. On the other hand, $T_{\alpha,PP}$ values and α -relaxation intensities of PGMA/CNF composites decreased slightly with increasing CNF content, which is believed to originate from the presence of β -form PP crystals and CNFs. The decrement of $T_{\alpha,PP}$ value can be supported by the study that the increment of the β -form content contributes to increasing the fracture toughness of PP.^[38] Consistently, $T_{\beta,PP}$ (16.1°C) and $T_{\alpha,PP}$ (83.0°C) of PP/CNF10 composite were found to be higher and lower than the values (14.8 and 88.0°C) of neat PP, respectively. In addition, the α -relaxation intensity of PP/CNF10 composite is weaker than that of neat PP. The high $T_{\beta,PP}$ and low $T_{\alpha,PP}$ with lowered α -relaxation intensity of PP/CNF10 composite, compared with neat PP, are related to the limited amorphous chain mobility and the β -form crystals of PP owing to the added rigid CNFs, respectively.

The impact strength test was carried out to determine the effect of CNF loading content on the toughness of PGMA/CNF composites. The toughness of composites is

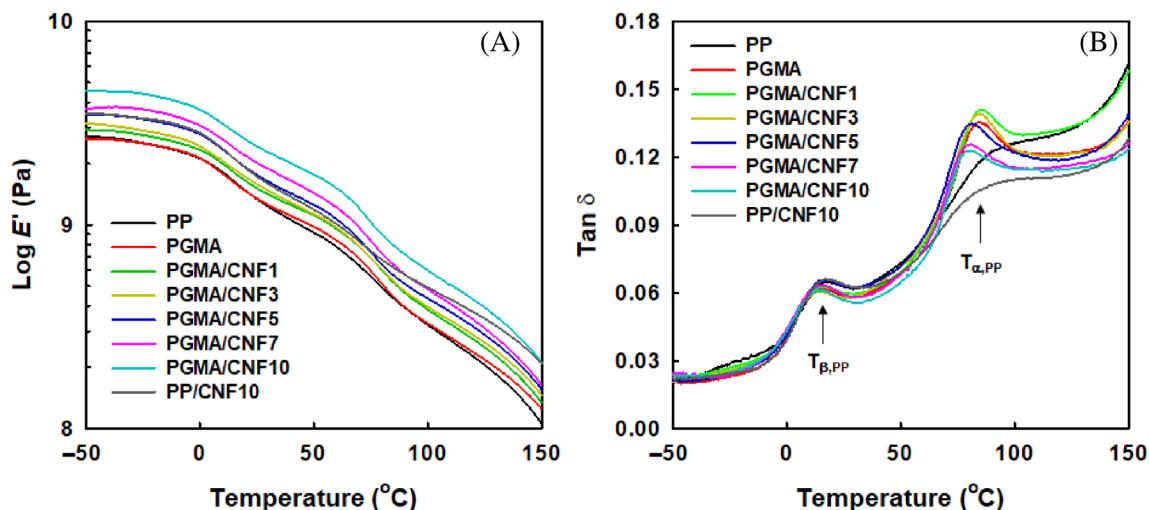


FIGURE 10 Temperature-dependent changes of (A) storage modulus (E') and (B) loss tangent ($\tan \delta$) of PP, PGMA, and their composites with different CNF contents

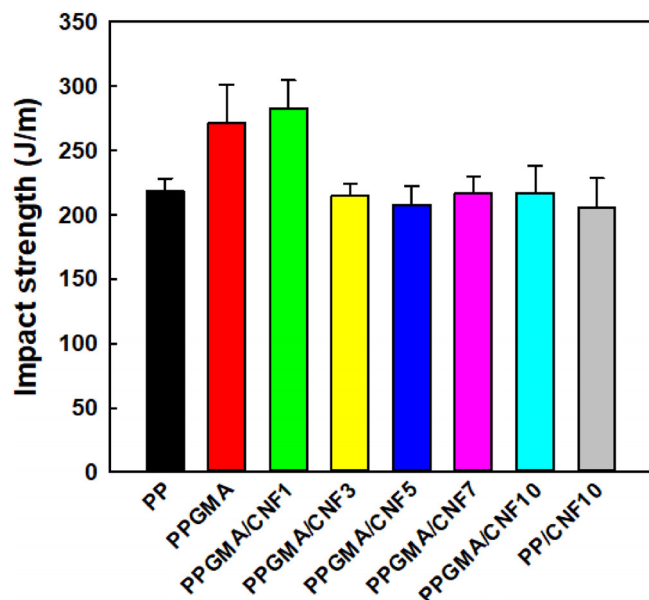


FIGURE 11 Izod impact strength of PP, PGMA, and their composites with different CNF contents

influenced by the reinforcement, the matrix, and their interfacial bonding strength. The Izod impact strength of PP, PGMA, and their composites are summarized in Figure 11. Neat PP was ductile with a high impact strength of 218.8 ± 8.4 J/m. The impact strength of PGMA was found to be 271.3 ± 24.2 J/m and 24.0% higher than that of neat PP. The β -form PP is reported to have a high toughness for the small spherulites compared with the α -form PP.^[32,43,44] From the above DSC and XRD results, it is known that PGMA contains β -form crystals, which may explain the fact that the impact strength of PGMA is higher than that of PP. For PGMA/

CNF composites, the impact strength increased to 282.8 ± 17.7 J/m at 1 wt% CNF loading, but it decreased to ~ 214 J/m at higher CNF loadings of 3–10 wt%. The presence of maximum impact strength for PGMA/CNF composites at 1 wt% CNF loading is owing to the trade-off effect between the enhanced compatibility of CNFs and PGMA by specific interactions and the increased rigidity by the addition of CNFs. For comparison, the impact strengths of PP/CNF1 (217.2 ± 2.8 J/m) and PP/CNF10 (205.3 ± 20.1 J/m) composites were measured to be lower than those of PGMA/CNF1 ($\sim 282.8 \pm 17.7$ J/m) and PGMA/CNF10 (216.4 ± 12.7 J/m), respectively. It was mostly due to the poor compatibility between the PP matrix and CNFs.

4 | CONCLUSIONS

In summary, to achieve thermoplastic polymer-based composites with enhanced thermal and mechanical properties, PGMA was manufactured by grafting reaction of PP with GMA and styrene via in situ melt-mixing and it was then reinforced by 1–10 wt% CNF loading content through masterbatch-based melt-compounding. The XPS, FT-IR, and EDS data confirmed the successful fabrication of PGMA as well as the strong hydrogen-bonding interactions between the CNFs and the PGMA matrix in the composites. The melt-rheological analysis revealed the existence of chemical reactions between the hydroxy group of CNFs and the glycidyl group of PGMA in the composites. The SEM images of fracture surfaces of PGMA/CNF composites confirmed the improved interfacial adhesion between CNFs and the PGMA matrix, compared with PP/CNF composite. Accordingly, the thermal

decomposition temperature ($T_{d,PP}$) of the PGMA matrix of PGMA/CNF composites of 458.5–461.5°C were found to be far higher than neat PGMA (435.1°C) or even neat PP (453.6°C). The DSC thermograms demonstrated that CNFs contribute to accelerating the crystallization of PGMA in the composites. In addition, XRD patterns showed that the β -form PP crystals were developed in neat PGMA and PGMA/CNF composites, except neat PP with only α -form crystals. The DMA analysis exhibited that the storage moduli of PGMA/CNF composites increased noticeably with the increment of CNF content. On the other hand, PGMA composite with 1 wt% CNF loading was found to have a maximum impact strength (~ 282.8 J/m), which is far higher than that (~ 218.8 J/m) of neat PP by 29.2%, owing to the trade-off effect between the enhanced compatibility of CNFs and PGMA by specific interactions and the increased rigidity by the addition of CNFs. Overall, it is valid to contend that the improved thermal and mechanical properties of PGMA/CNF composites could be attained by the specific interactions as well as the chemical reactions between the GMA group grafted on PP and the hydroxyl group of CNFs. It is thus believed that PGMA/CNF composites can be applied as eco-friendly high-performance thermoplastic composite materials in various industrial sectors including automotive, packaging, building, construction, electrical, electronics, and so forth.

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CONFLICT OF INTEREST

All the authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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